Modelling Limit Order Books

A Markovian approach using Diffusion Models

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What are Limit Order Books?

- **Limit Order Books** (LOBs) are records of aggregated outstanding **limit orders** (orders submitted at a pre-determined price and quantity).

Message book

Time(sec)	Price($)	Volume	Event Type	Direction
34203.011926972	585.68	18	execution	ask
34203.011926973	585.69	16	execution	ask
...	...	...	...	...
34203.011988208	585.74	18	cancellation	ask
34203.011990228	585.75	4	cancellation	ask
...	...	...	...	...
34203.012050158	585.70	66	execution	bid
34203.012287906	585.45	18	submission	bid
34203.089491920	586.68	18	submission	ask

Figure: Example of L3 Order Book

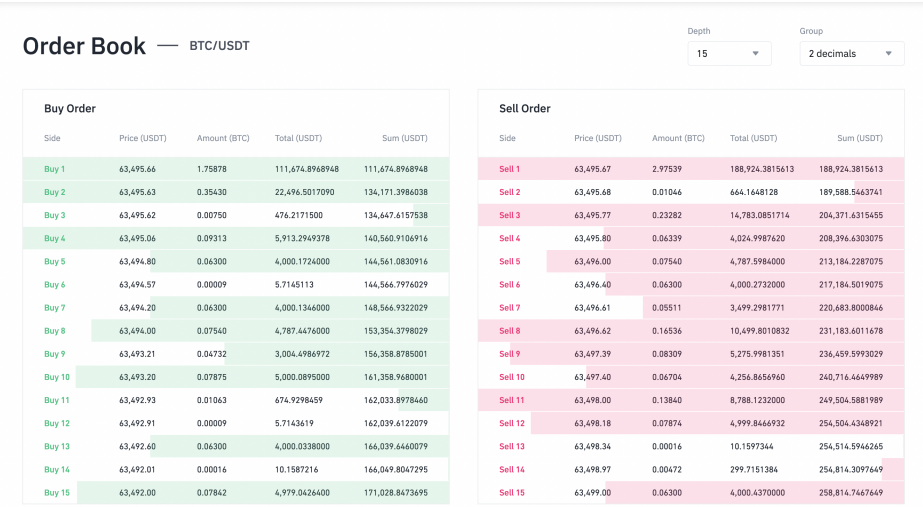


Figure: Example of L2 Order Book

Why model Limit Order Books?

- LOBs provide insight into price formation of securities at a granular level. Cont et al. [2] provide theoretical and empirical evidence that the "order-flow imbalance" has a linear relationship with high-frequency price changes.
- Interactive simulation environment [9] for new trading strategies without the need to risk capital. Adresses common issue of backtesting, where trading strategies perform well on historical data, but poorly on live data ("simulation-to-reality gap" Prata [5]). Microsoft released Order Book Simulator MarS in September 2024 using LMM (large market model)

How to model Limit Order Books?

Cont et al. [3] recently proposed general framework. Let $Q_p^-(t)$ and $Q_p^+(t)$ denote the amount of outstanding limit buy and sell orders at price p and time $t \in [0, T]$ respectively.

- Centered Order Book: Only consider the queue sizes d levels around the mid-price p_{mid}, as these are most informative, i.e. for $* \in \{-, +\}$

$$X^*(t) := \left(Q^*_{p_{\text{mid}}-d\delta}(t), \ldots, Q^*_{p_{\text{mid}}}(t), \ldots, Q^*_{p_{\text{mid}}+d\delta}(t)\right) \in \mathbb{N}^{2d+1}.$$

Define the operator $\operatorname{supp}(X) := \{i \in \{p_{\text{mid}} - d\delta, \ldots, p_{\text{mid}} + d\delta\} : x_i > 0\}$.

(i) $b(X(t)) := \sup\{\operatorname{supp}(X^-(t))\}$ is called the **bid price**.

(ii) $a(X(t)) = \inf\{\operatorname{supp}(X^+(t))\}$ is called the **ask price**.

Order book representation

- Dynamics: Incoming orders of size $z > 0$ at price $i \in \{p_{\text{mid}} - d\delta, \ldots, p_{\text{mid}} + d\delta\}$

$$\begin{cases} (X^-, X^+) \mapsto (X^- + ze_i, X^+) & \text{limit buy} \\ (X^-, X^+) \mapsto (X^-, X^+ + ze_i) & \text{limit sell} \\ (X^-, X^+) \mapsto (X^- - ze_i, X^+) & \text{cancellation buy} \\ (X^-, X^+) \mapsto (X^-, X^+ - ze_i) & \text{cancellation sell} \end{cases}$$

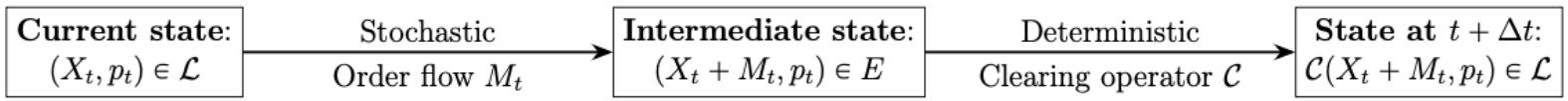
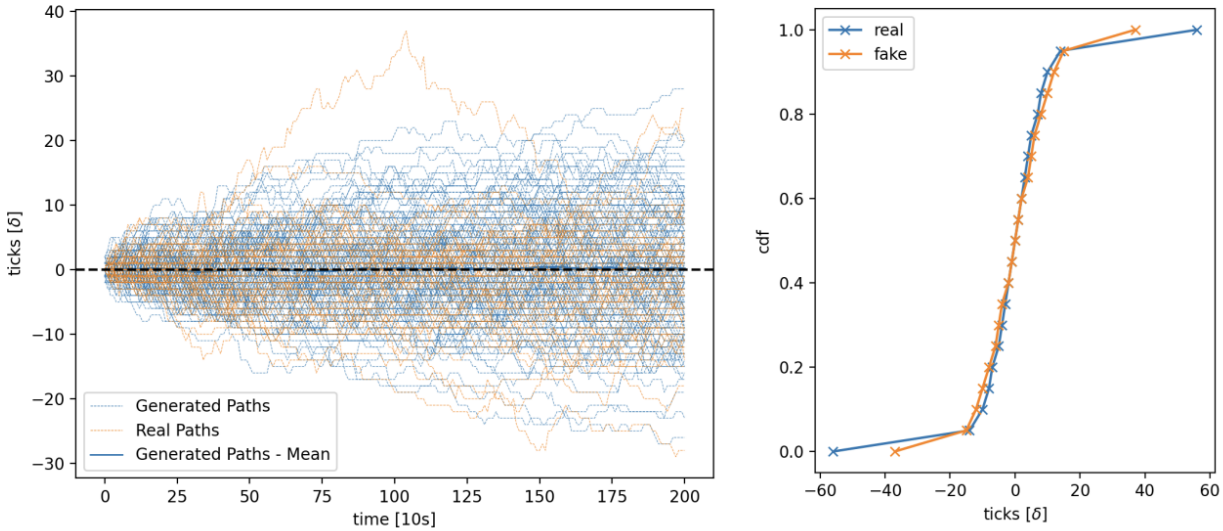


Figure 3: Limit Order Book Decomposition

Machine-learning Approaches

- Cont et al. (2023) [4] use a GAN to learn order-book transitions over a 10 second period.



(a) Real and fake price paths (in ticks).

(b) CDF of $\Delta p_{mid}(200)$ in ticks.

Figure 7: Price paths and corresponding percentiles of terminal prices for real and fake data.

Diffusion models for Time Series

- Recent study by Meijer (2023) [8] suggests that diffusion models are also able to handle time-series data, producing state-of-the-art results on GluontTS benchmark time-series data.

Paper	Name	Method	Input Data	AR	Gen. Method	Rev. Model	Task	Horizon
[64] (3.1)	TimeGrad	Cond. Model	⬅	🔁	DDPM	Noise	🔮	Short
[62] (3.2)	ScoreGrad	Cond. Model	⬅	🔁	SDE	Score	🔮	Short
[65] (3.3)	CSDI	Cond. Model	↔	🔧	DDPM	Noise	🩹	Short
[66] (3.4)	DSPD	Cond. Model	↔	🔧	DDPM	Noise	🩹	Short
[66] (3.4)	CSPD	Cond. Model	↔	🔧	SDE	Noise	🩹	Short
[67] (3.5)	D^3VAE	Cond. Model	⬅	🔧	Coupled DPM	BVAE	🔮	Short
[68] (3.6)	TDSTF	Cond. Model	⬅	🔧	DDPM	Noise	🔮	Short
[69] (3.7)	$SSSD^{S4}$	Cond. Model	↔	🔧	DDPM	Noise	🩹	Long
[17] (3.8)	DiffLoad	Cond. Model	⬅	🔁	DDPM	Noise	🔮	Short
[70] (3.9)	TimeDiff	Cond. Model	⬅ + Extras	🔧	DDPM	Data	🔮	Long
[71] (3.10)	TSDiff	Guidance	⬅	🔧	DDPM	Noise	🔮	Short

Figure: Diffusion models for time-series data

Diffusion Models

- **Approach:** Denoising diffusion probabilistic model (DDPM) proposed by Ho et al. [6]. Given data distribution Q and sample data $x \sim Q$, iteratively perturb x with Gaussian noise.

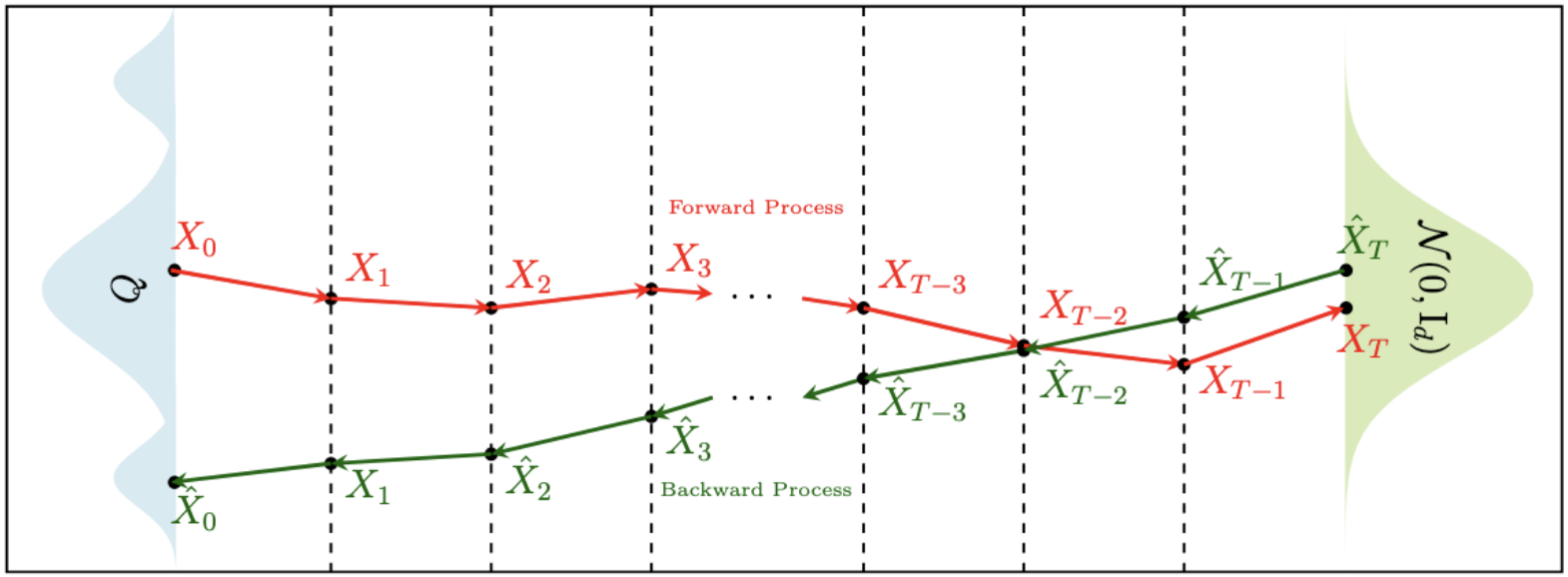


Figure 4: Forward and Backward process of Diffusion model

DDPM Approach

- For given variance schedule $(\beta_t)_{t=1}^T$ and i.i.d. $\varepsilon_1, \ldots, \varepsilon_t \sim \mathcal{N}(0, \mathrm{I}_d)$, recursively define the noising/forward process for data point $x \sim Q$, as $x_0 := x$ and

$$x_t := \sqrt{1-\beta_t} x_{t-1} + \sqrt{\beta_t} \varepsilon_t.$$

- **Objective:** Learn the denoising/backward process $p_\theta(x_{t-1} \mid x_t)$ and thus the joint distribution

$$p_\theta(x_0, \ldots, x_T) = p_\theta(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t).$$

Assume $p_\theta(x_T) = \mathcal{N}(0, \mathrm{I}_d)$. $p_\theta(x_t \mid x_{t-1}) = \mathcal{N}(\mu_\theta$, which is learned through minimizing negative log-likelihood $\mathbb{E}_q[-\log p_\theta(x)]$.

Derivation of DDPM in a nutshell

$$
\begin{aligned}
\mathbb{E}_q[-\log p_\theta(x_0)] &\leq -\mathbb{E}_{q(x_0,x_{1:T})}\left[\log \frac{p_\theta(x_0,x_{1:T})}{q(x_{1:T}\mid x_0)}\right] \\
&= \underbrace{\mathbb{E}_{q(x_{0:T})}\left[-\log p_\theta\left(x_0|x_1\right)\right]}_{\text{Reconstruction term}} \\
&+ \underbrace{\sum_{t=2}^{T}\mathbb{E}_{q(x_0,x_t)}\left[\mathbf{d}_{\mathsf{KL}}\left(q(x_{t-1}\mid x_0,x_t)\|p_\theta(x_{t-1}|x_t)\right)\right]}_{\text{Denoising-matching term (*)}} \\
&+ \underbrace{\mathbb{E}_{q(x_0,x_T)}\left[\mathbf{d}_{\mathsf{KL}}\left(q(x_T|x_0)\|p_\theta(x_T)\right)\right]}_{\text{Prior-matching term}}
\end{aligned}
$$

- $(*) = \sum_{t=2}^{T}\alpha_t\mathbb{E}\left[\left\|\varepsilon_t-\varepsilon_\theta\left(x_t,t\right)\right\|^2\right]$ (Loss function in DDPM).

For details: Unified perspective on Diffusion Models [7].

Modelling Limit Order Book Snapshots

Uniformly sample a centered LOB process $(\bar{O}_t)_{t\in[0,T]}$ with frequency Δt, i.e. $O_k := \bar{O}_{k\Delta t}$ for $k \in \mathbb{N}$. We are interested in learning the distribution of the snapshot O_{k+1} given the information up to time k, which we encode as a state variable $S_k \in \mathbb{R}^s$. In other words,

$$O_{k+1} \sim \mathbb{P}_q(o_{k+1} \mid S_k).$$

The state variable include information like macroeconomic news or temporal data. Consider a Markovian setting, where $S_k = O_k$.

Diffusion Model for LOB Modelling

Model Architecture: TimeGrad proposed by Rasul et al. (2021) [1].

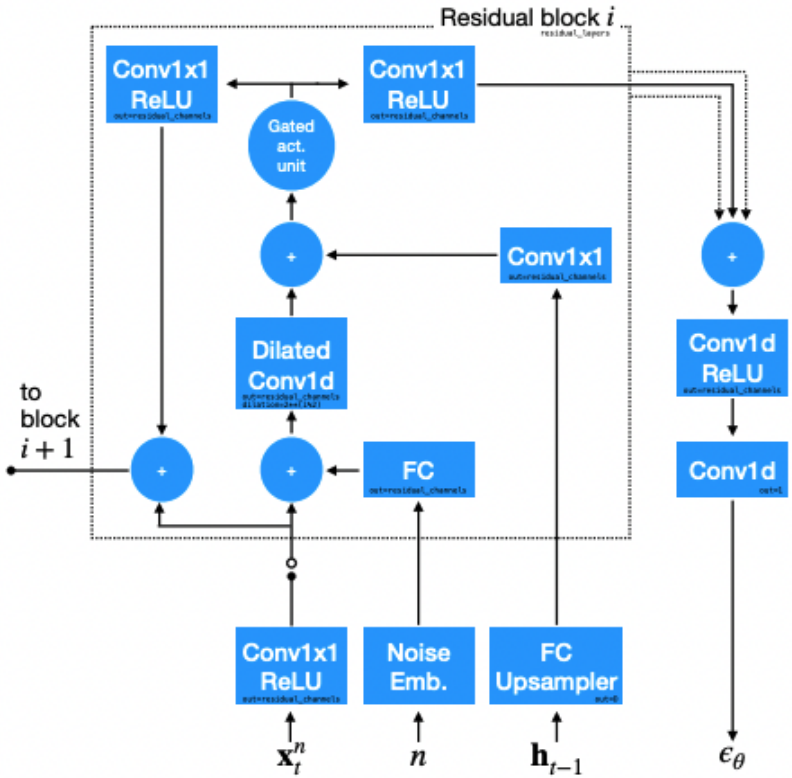


Figure: Model Architecture of TimeGrad

Experimental Setup

- **Training Dataset**: BTC/USDT data from 10/10/2023-10/15/2023 ($\approx$ 100,000 data points).
- **Validation Dataset**: 10/16/2023-10/17/2023.
- **Test Dataset**: 10/18/2023-10/19/2023.

Variance schedule β_t	Cosine
Diffusion steps T	80
Loss type	L2
Learning Rate	$1e^{-3}$
Epochs	30
Number of batches per epoch	100
Batch size	64

Table: Hyperparameters

Generated vs Actual Price Paths

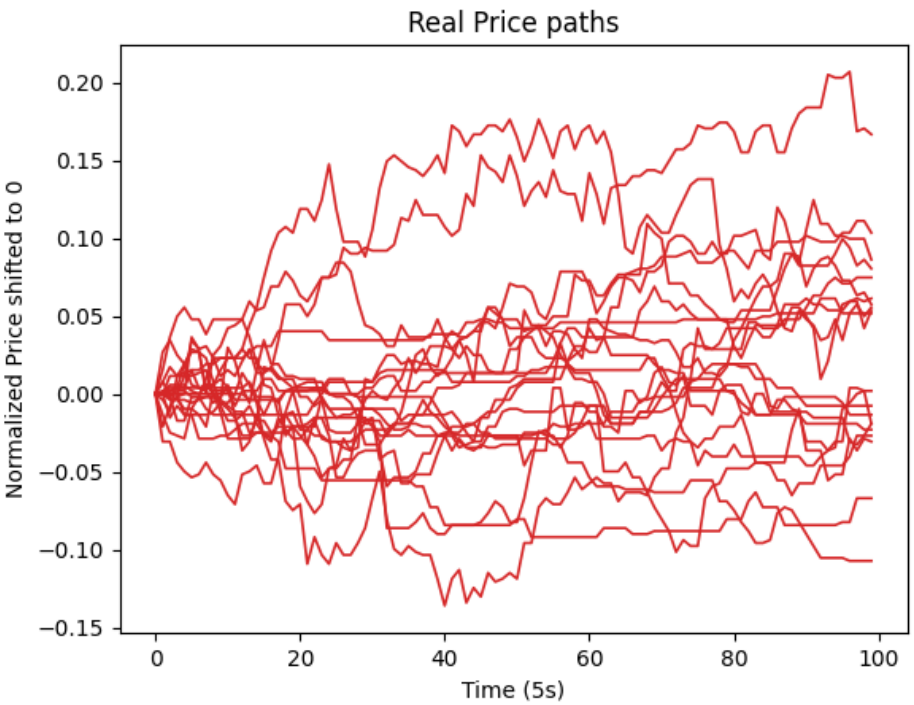


Figure: Actual price paths

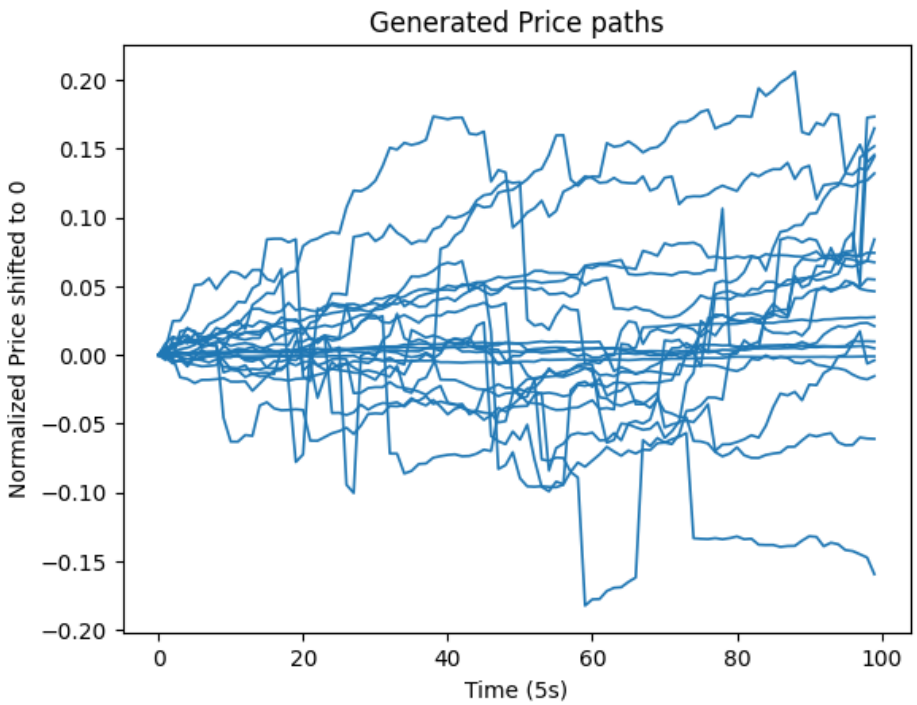


Figure: Generated price paths

Evaluation

Stylized facts are statistical properties that are common across a wide range of instruments, markets and time periods.

(1) Distribution of log-returns: Log-returns defined as

$$r(t, \Delta t) = \log\left(\frac{p_{\text{mid}}(t + \Delta t)}{p_{\text{mid}}(t)}\right)$$

are normally distributed over longer time-scales Δt. The distribution of smaller Δt typically display heavy tails and have high kurtosis.

Time-scale (Δt)	True Return Kurtosis	Generated Return Kurtosis
5s	483	191
30s	470	31
1min	632	14
5min	614	2
30min	56	N/A
1h	14	N/A

Table: Comparison of Kurtosis for true and generated returns

Stylized fact 1: Distribution of log-Returns

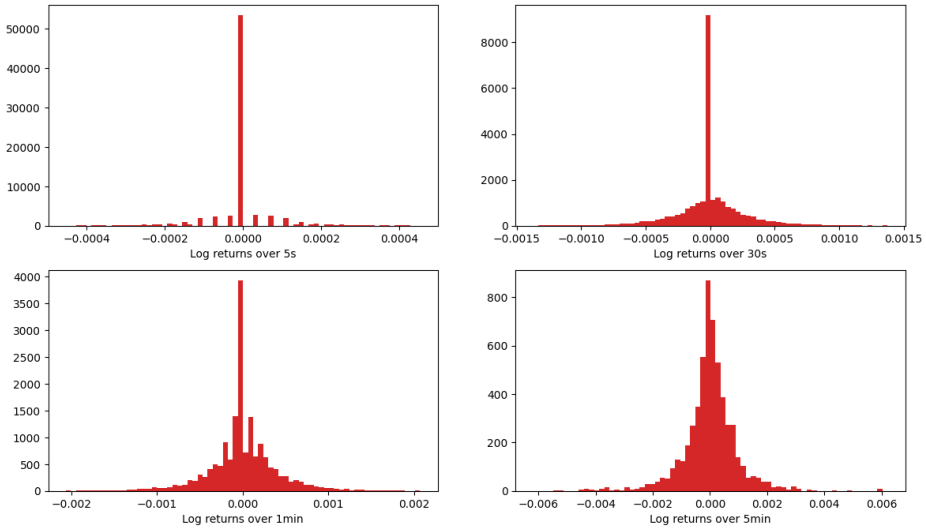


Figure: Actual log-returns

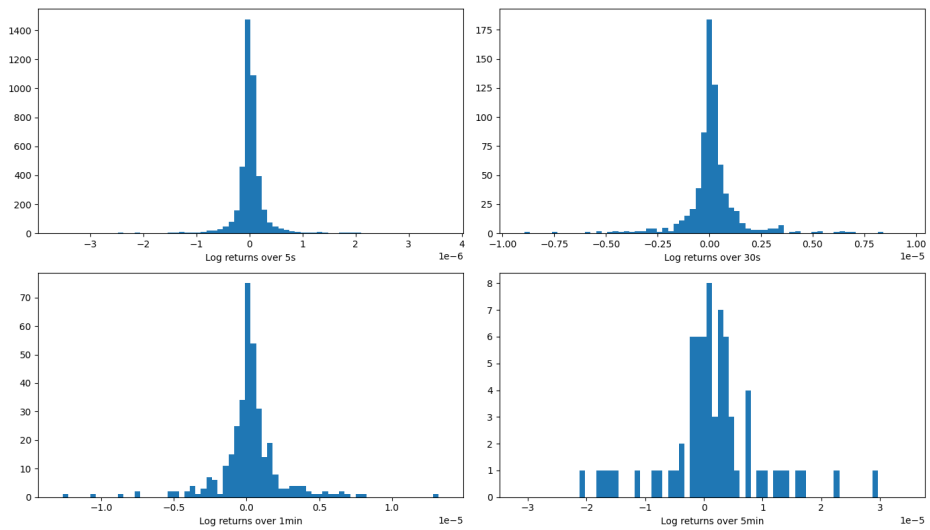


Figure: Generated log-returns

Stylized fact 2: Absence of autocorrelation

Empirically, the autocorrelation rapidly decays to 0.

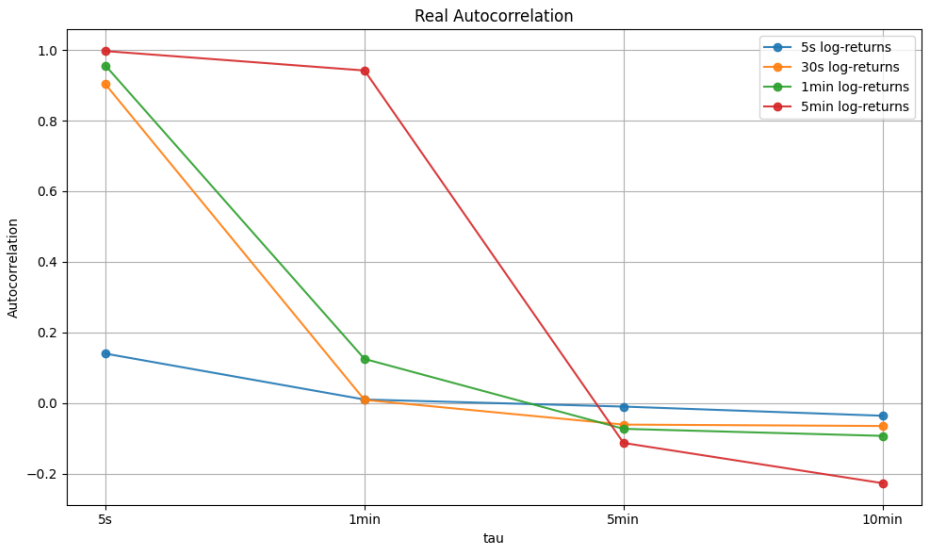


Figure: Real Autocorrelation

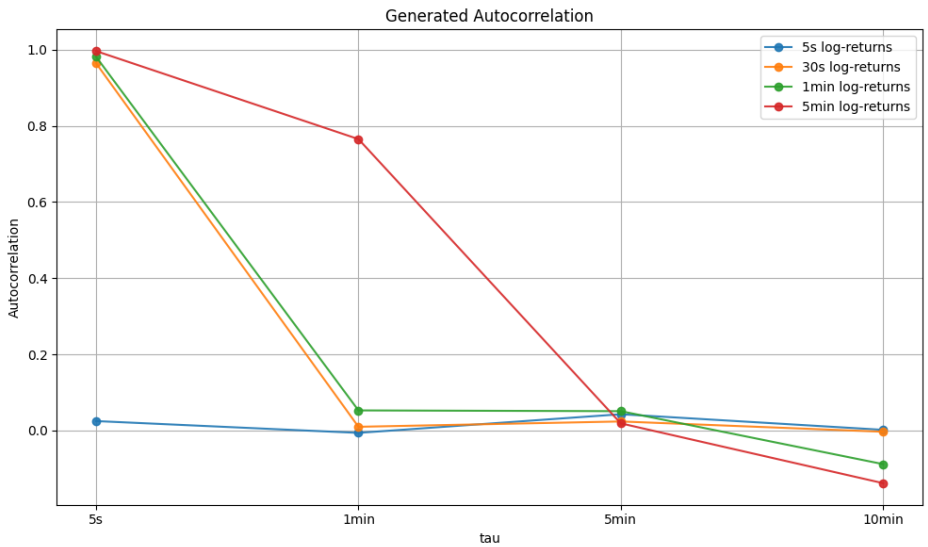


Figure: Generated Autocorrelation

Stylized fact 3: Volatility Clustering

Volatility clustering and implies that periods of high and low volatility typically persist for period of time. The autocorrelation function of absolute returns does not vanish as quick as the linear autocorrelation.

$$C(\tau) = \mathrm{corr}(|r(t, \Delta t)|, |r(t + \tau, \Delta t)|).$$

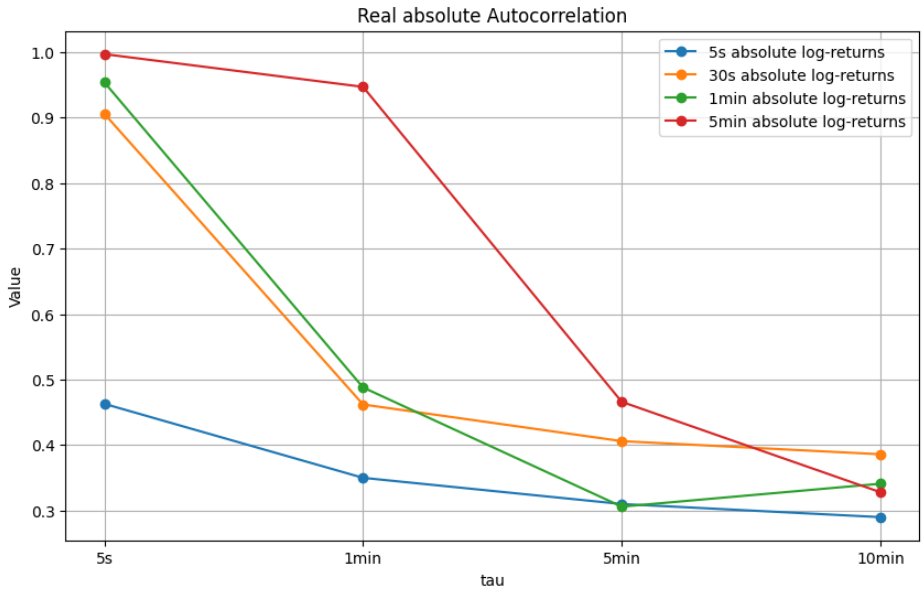


Figure: Real abs Autocorrelation

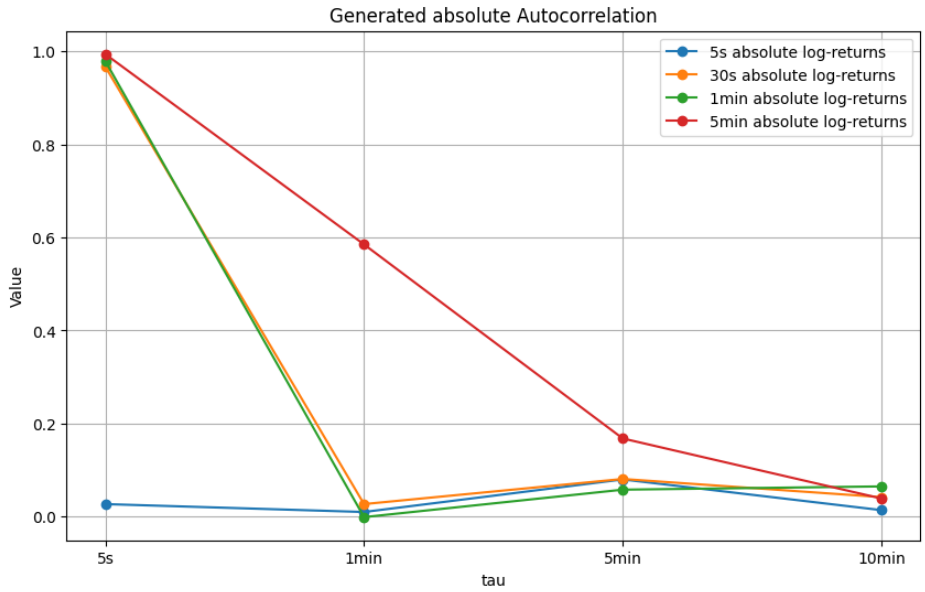


Figure: Generated abs Autocorrelation

Possible Extensions

- Speed up inference and generate time series for entire period instead of auto-regressively (TimeDiff, 2023)
- Beyond Markovian assumption and multi-modal approach to encode additional information into state S_k
- Convergence bounds for different variance schedules

[1] Kashif Rasul et al. *Autoregressive Denoising Diffusion Models for Multivariate Probabilistic Time Series Forecasting*. 2021. URL: https://arxiv.org/pdf/2101.12072.pdf.

[2] R. Cont, A. Kukanov, and S. Stoikov. "The Price Impact of Order Book Events". In: *Journal of Financial Econometrics* 12.1 (June 2013), pp. 47–88. ISSN: 1479-8417. DOI: 10.1093/jjfinec/nbt003. URL: http://dx.doi.org/10.1093/jjfinec/nbt003.

[3] Rama Cont. *A mathematical framework for modelling order book dynamics*. 2023. URL: https://arxiv.org/pdf/2302.01169.pdf.

[4] Rama Cont et al. *Limit Order Book Simulation with Generative Adversarial Networks*. 2023. URL: https://ssrn.com/abstract=4512356.

[5] Matteo Prata et. *LOB-Based Deep Learning Models for Stock Price Trend Prediction: A Benchmark Study*. 2023. URL: https://arxiv.org/pdf/2308.01915.pdf.

[6] Jonathan Ho, Ajay Jain, and Pieter Abbeel. *Denoising Diffusion Probabilistic Models*. 2020. arXiv: 2006.11239 [cs.LG]. URL: https://arxiv.org/abs/2006.11239.

[7] Calvin Luo. *Understanding Diffusion Models: A Unified Perspective*. 2022. arXiv: 2208.11970 [cs.LG]. URL: https://arxiv.org/abs/2208.11970.

[8] Caspar Meijer and Lydia Y. Chen. *The Rise of Diffusion Models in Time-Series Forecasting*. 2024. arXiv: 2401.03006 [cs.LG]. URL: https://arxiv.org/abs/2401.03006.

[9] Svitlana Vyetrenko et al. *Get Real: Realism Metrics for Robust Limit Order Book Market Simulations*. 2019. arXiv: 1912.04941 [q-fin.TR].